

Sturgeon Bay

Shoreline length: 34 km



Shore Types

This SMU includes Clarks Bay and Sturgeon Bay, sheltered behind Panmure Island. Main shore types are low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
71	2	0	1	26	0	0	0



Thornton Point and Clarks Bay inside Sturgeon Bay

Governance, Infrastructure, and Economy

Sturgeon Bay falls almost entirely within the Rural Municipality of Three Rivers; approximately 10% of the shore of the estuary is unincorporated.

Key infrastructure includes the Sturgeon wharf and Old Wharf Road, Route 17, Pembroke Rd and St. Andrews Point Road. Coastal economic drivers include Sturgeon Wharf which supports an active fishery.

Local and legacy land use constraints include several subdivisions with small lots on unpaved roads running parallel or perpendicular to the coast.

Planning and Policy Framework

Where provincial rules apply, land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a

coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted. Within Three Rivers, the Town's municipal rules apply. Development on the coast is informed by the Environmental Protection Act, including a 15m buffer, with no additional building setbacks aside from the buffer. The Official Plan includes specific guidance permitting armouring, though such provisions are secondary to any provincial standards, and includes some policies regarding expansion of buildings within the sea level rise overlay. There are no specific flood elevations or vertical setbacks established, but the Official Plan does include specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Sturgeon Bay falls entirely under municipal planning authority of Three Rivers. Coastal development is informed by provincial legislation, including the Environmental Protection Act. The Three Rivers Official Plan includes policies regarding expansion of buildings in areas affected by sea level rise. No specific flood elevations or vertical setbacks are established.

Official plan includes specific erosion-based setbacks, with the possibility of reduced setbacks under certain conditions. Hazards are assessed at the subdivision stage.

Economic drivers include Sturgeon Wharf which supports an active fishery. Local land use constraints include several subdivisions with small lots on unpaved roads running parallel or perpendicular to the coast.

Coastal Flood and Erosion Risk

While low plains and wetland shores are vulnerable to increasing flood hazards, infrastructure risk is low. Erosion hazard is limited due to the sheltered nature of the area except along the more exposed headlands. The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential shoreline change or salt marsh migration to future time horizons.

Number of at-risk buildings [N. per km shoreline] in SMU

	Present	0.3m SLR (2050)	0.6m SLR (2080)	1.2m SLR (2130)
Flood risk	6 [<1/km]	8 [<1/km]	11 [<1/km]	13 [<1/km]
Erosion risk	N/A	10 [<1/km]	15 [<1/km]	23 [1/km]

Sediment transport - The estimated net sediment transport rate averaged across the SMU is moderate. It feeds a few beaches extending into the bay such as Thornton Point, Meadowfoot Point and Wight Point.

Proposed SMP Strategies

The primary strategy for Sturgeon Bay is No Active Intervention for the short-term, followed by MR. The need for MR is primarily driven by flood risk.

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
Proposed SMU strategy	NAI	MR	MR
Rationale	Mid- to long-term MR will reduce risk to the limited number of vulnerable assets while preserving natural shorelines, including wetlands and low plains with increasing flood hazard.		



LEGEND - Shoreline Management Unit - Net Longshore Sediment Transport - High Hazard Area - Flood Extent - Present Day (100-year Return) - Flood Extent - 1.2m SLR (100-year Return) - Property Boundary - Building > 90ft²	 0 170 340 680 Metres Coordinate System: NAD 83 CSRS PEI Double Stereographic NOTES:		COASTAL EROSION AND FLOOD HAZARD Shoreline Management Unit: Sturgeon Bay	<i>Prince Edward Island</i> Île-du-Prince-Édouard
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